

Music Composition with Interactive Genetic Algorithms

Sokolo D. Matsolo

Computer Science and Software Engineering
University of Johannesburg
Johannesburg, RSA
222206953@student.uj.ac.za

Abstract— Recently, scientists studying artificial intelligence have become interested in the intersection of computer science and music. Computers have consistently showcased their problem-solving prowess, and the domain of music is no exception to their impact. Although users are able to create their own music, the vast number of possible music combinations makes organized compilation challenging making it difficult for average users to create emotionally engaging musical patterns. Using Interactive Genetic Algorithms (IGAs), which are renowned for their iterative evolutionary optimization capabilities, is a promising way to customize music to each user's unique emotional preferences. This study explores the intersection of computer science and musical creativity, demonstrating how IGAs can enable users to create music that perfectly complements their moods and tastes. The paper is structured into three key sections: firstly, it offers a concise overview of the foundational concepts and components of IGAs; secondly, it presents the formal definition of the music, setting the stage for the problem-solving endeavor; and thirdly, it extensively delves into the practical implementation of GAs to tackle the interactive evolving music, constituting the primary focus of this research. Within the implementation phase, the paper explores selection functions, namely the roulette wheel and elitism selection methods. Furthermore, the program's robustness is tested across varying mutation rates and a configuration of a single-point crossover, revealing intriguing insights into their impact on the algorithm's performance.

Keywords- *Evolutionary Computation; Interactive Genetic Algorithm; Music Composition*

I. INTRODUCTION

In our daily lives, we are exposed to a variety of multimedia content, but music is one that we mostly find enjoyable. The practice of listening to music has the power to change how we feel, and music therapy makes use of this power. Recent developments in computer science have made it possible for everyone to manipulate music on computers. However, inexperienced composers face many challenges when it comes to writing their own music. A foundation of musical experiences and knowledge, including familiarity with melodies and chord progressions, is required for such projects [1].

Swarm intelligence, neural networks (NN), and evolutionary computation (EC) are some of the fundamental

building blocks of computational intelligence (CI). These concepts have been used to create collaborative techniques and implemented in user-tested products. But in terms of creativity, human beings continue to perform best. This is due to the fact that creativity frequently depends on subjective opinions and feelings when evaluating our own creations [4].

IEC is an innovative methodology in research that stands out. It is centered on people actively participating in the process of determining whether art is favorable. A specific type of GA is the dominant evolutionary strategy in IEC systems. An improved version is called Interactive Genetic Algorithms (IGAs). Earlier studies have looked into its use in the field of music composition. Alternative GAs with the goal of improving the efficiency of solution searches are becoming more and more popular, according to recent IEC studies. These alternatives cover a range of methods, such as an algorithmic fitness function that makes use of the normalized Compression Distance, a user-intervention technique that allows direct control over musical melodies by adjusting note keys, and a historical emphasis on melody generation that places a premium on chord progressions [3, 4, 6]. Interestingly, there is still an obvious absence of study on direct user engagement, which suggests that more research is urgently needed in this area, even in spite of earlier suggestions from other IEC studies.

In this study, a genetic algorithm is utilized as it allows for the generation of several songs. High-quality compositions can be selected by applying the proper fitness function to this generated music and user interaction. This strategy relieves composers of the burden by releasing them from the need to write original music for each occasion. The produced music is used as useful samples for music databases, which helps with system training, and as background music in songs, reels, and videos.

II. INTERACTIVE GENETIC ALGORITHM

A. Genetic Algorithm and Interactive Genetic Algorithm

Genetic algorithms (GAs) are computer techniques that seek favorable solutions from a large set of possible replies.

John Holland, his students, and colleagues at the University of Michigan pioneered the concepts of genetic algorithms in the 1960s. These computational approaches are inspired by natural evolution principles, including notions such as survival of the fittest, reproduction, and mutation [8]. These ideas are well suited to tackling a wide range of practical difficulties across several areas. Optimization, automated coding, machine intelligence, economics, immune systems, population genetics, and social systems are some of the uses for genetic algorithms.

GAs begin with a population of potential solutions. The goal is to create a new population from the solutions in the previous one in order to achieve a more improved population. The fitness of solutions to contribute to the development of new offspring determines their selection. Solutions that are more suitable have a better chance of being replicated. This iterative approach is repeated until a specific condition is fulfilled [8].

The Interactive Evolutionary Algorithm (IGA) is an evolutionary algorithm that optimizes subjective artificial evaluations [10]. Subjective elements such as human preferences, intuition, emotions, and psychological traits are factored into the appraisal of new species during the evolutionary process. This technique eliminates the drawbacks of unpredictability and passivity that might arise in optimization procedures, hence overcoming the challenge of implicit performance indicator optimization [3]. The fitness of the evolving species is determined by human judgment, which separates IGA from traditional genetic algorithms.

B. Fundamental components of IGAs

IGAs, like the majority of GA approaches, rely on the following elements: chromosomal representation and populations, fitness-based selection, crossover to produce new offspring, and random mutation of the new offspring.

Initialization:

The GAs set up the search space by generating an initial population. In the majority of GA implementations, IGAs included, individuals within this initial population have their attributes distributed at random and uniformly within predefined upper and lower bounds.

Chromosome Representation:

Chromosomes are the basic building blocks of GAs, which indicate potential solutions to a search space. These chromosomes may be encoded using binary, permutation, value (float), or tree encodings, among other methods. The type of problem and the characteristics of the entities involved determine the best encoding technique. For instance, it's typical to display numerous people at once when dealing with people who represent photos and graphics so that consumers can compare them. However, since it is impractical for consumers to evaluate numerous people at once, individuals are shown sequentially in time-sequential media material. Each GA

(specifically IGA) solution must be presented in its true form, or phenotype, for human evaluation to ensure an accurate evaluation [5].

Fitness Function:

GAs require the existence of a fitness function that is in charge of presenting each chromosome in the present population with a score. With the aid of this function, the GA can evaluate the strength and effectiveness of the encoded solutions in relation to the particular problem at hand [5]. This review in IGAs is accomplished through user ratings or selections, where users opportunistically assess the supplied individuals.

Selection:

The selection process in GAs is crucial for creating the next generation. It relies on the fitness of chromosomes, with those having higher values (i.e., being fitter) having a greater chance of being chosen for reproduction. Conversely, chromosomes with lower fitness values are discarded. This process mimics the survival of organisms in nature and incorporates an element of randomness. GAs use a variety of selection techniques, including random selection, rank selection, and proportional selection. Elitism is another way in which GAs, particularly IGAs, might improve their performance. Several of the highest-scoring chromosomes from the current generation are copied to the new population as part of elitism. This makes sure that the greatest solutions thus far are maintained while still producing the rest of the population.

Crossover:

One of the GAs' reproduction operators, crossover, is essential for producing offspring for the following generation. This process tries to produce children for the following generation who will carry qualities from both parents' chromosomes. A random point is chosen for the crossover procedure, and the sequences that occur before and after that point are switched between the two chromosomes to create two unique offspring [8]. Consider the parents listed below and the crossing point at index 3 as an example:

Parent 1: 0 1 1 | 0 1 0 1

Parent 2: 1 0 1 | 1 0 1 0

Offspring 1: 0 1 1 1 0 1 0

Offspring 2: 1 0 1 0 1 0 1

Both Offspring 1 and Offspring 2 inherit bits 1-3 from one parent's left side and the remaining bits from the other parent's right side during the crossover.

Mutation:

Mutation represents another operation within a genetic algorithm's reproduction. Mutation is conducted following the crossover process as to avoid confining all solutions within the population to a local optimum of the solved problem. It alters the new offspring by flipping its genetic material. Returning to the crossover example, consider a mutation occurring at index 2 in the genetic sequence of offspring 2:

Offspring 2: 1 0 1 0 1 0 1

Mutated Offspring 2: 1 1 1 0 1 0 1

Algorithm 1 Basic Genetic Algorithm

- 1: **Start:** Generate a population of N chromosomes randomly.
 - 2: **Fitness:** Evaluate the fitness of each chromosome.
 - 3: **Create a new population:**
 - a. **Selection:** Apply the selection technique to pick two chromosomes from the population
 - b. **Crossover:** Apply crossover operations to the selected pair of chromosomes
 - c. **Mutation:** Introduce mutations to offspring obtained in the previous step.
 - 7: **Replace:** Swap the existing population with the newly created population.
 - 8: **Test:** Examine if the termination condition is met. If it is, conclude the process; otherwise, return the top-performing solution from the current population and proceed to Step 2.
-

III. MUSIC

A. Basic Music Theory

The fundamentals of music theory outline the elements of harmony, melody, and rhythm. Its description of compositional aspects includes musical structure, rhythm, musical notes, chords, key signatures, intervals, scales, and other factors. Additionally, it recognizes many musical qualities such as dynamics, pitch, duration, timbre, and other elements. These fundamental characteristics of sound have reflected important links across the fields of mathematics, physics, and music.

Pitch:

The fundamental idea in music theory is pitch, which serves as the foundation for the creation of music. It embodies our sensory awareness of the vibrations made by musical instruments or voices, and it is essential in establishing how high or low musical notes are perceived to be. To delve deeper into the intricacies of pitch, we must explore two essential components: intervals and scales.

An "interval" in music nomenclature denotes the disparity in pitch between any two notes. Taking this a step further, a "scale" offers an organized sequence of musical intervals. The diatonic scale, consisting of the seven notes C, D, E, F, G, A, and B, is one of the most used scales in Western European music. These notes appear repeatedly on all types of instruments, offering a unified and flexible foundation for musical composition. The "octave interval," which denotes the distance between any note and the note above it in the diatonic scale (for example, C to the next higher C), is a crucial idea. Because they lay the groundwork for the entire scale, understanding octaves is essential. Within an octave, the diatonic scale further subdivides into five whole-tone intervals and two half-tone intervals, thereby shaping the distinct character of this scale.

Timbre:

Timbre is how a musical note or sound sounds is perceived. It plays a crucial role in distinguishing various sounds in music and forms the foundation for the instrumentation phase in music composition. The timbre characteristics of different musical instruments are shaped by the harmonic components present in the sounds they produce. In music, there are no purely sinusoidal sounds; instead, one encounters intricate waveforms that arise from the presence of distinct harmonic elements in both amplitude and phase.

Duration:

Duration is defined as the length of time that a note or a pitch in music lasts. Notes and rests define the length and absence of sounds. In its simplest form, each note lasts for just one-twelfth of the time of the preceding note, for example:

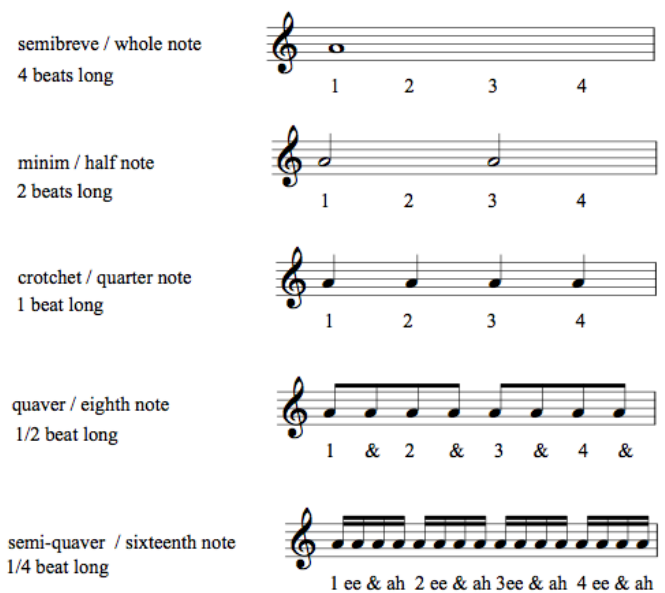


Figure 1: Durations of notes and rests.

B. Computational Approach to Music Generation

Music holds a significant role within culture, and as technology advances, the 20th and 21st centuries have emerged as eras characterized by the integration of computer-based music. The integration of music theory into computational music composition is essential for creating algorithmic systems that can produce aesthetically pleasing and musically coherent results. Understanding the principles of musical elements is instrumental in harnessing the creative potential of technology to produce innovative musical experiences [11]. Notably, composers like Iannis Xenakis, Lejaren Hiller, and John Cage embraced the possibilities of computer-based music production. These pioneers harnessed computers as tools for composition, resulting in innovative musical experiences.

Xenakis and Hiller, for instance, employed stochastic methods, and others have explored neural networks, rule-based systems, knowledge-based systems, evolutionary computation, and hybrid systems.

- Creating melodies
- Crafting harmonies
- Orchestrating a specified musical composition
- Adapting/innovating upon a provided musical theme

The first three tasks represent a comprehensive musical composition process, starting from initial melody concepts to the final musical composition. In contrast, adapting or innovating upon a provided musical theme is centered on variation and improvisation, and possesses unique characteristics while sharing some similarities with melody generation. Each of these musical problems may be seen as a search process where the goal is to find aesthetically attractive solutions within set limitations. These limitations are essential in guiding the creative process because of the almost infinite scope of the musical cosmos.

When it comes to melody generation, one strategy involves confining the exploration within a specific musical scale, thereby restricting the choice of available pitches or assigning them hierarchical significance. For instance, in the pursuit of serial themes, composers might ground their compositions in the chromatic scale and abide by the rules of serialism. Harmony generation, on the other hand, leans toward determinism, permitting the application of harmonization rules and counterpoint techniques. As for instrumentation, it varies with the musical genre and artistic preferences, guided by considerations such as the limitations of musical instruments and established knowledge regarding effective instrument combinations.

IV. IMPLEMENTATION OF MUSIC USING IGAs

A. Proposed Method

While existing models utilize a variety of fitness functions, including machine learning-based, automotive fitness functions, and knowledge-based approaches, Akanksha et al. proposed a model that predominantly relies on user ratings for fitness evaluation. The study demonstrated that music is a highly subjective domain, and conventional fitness functions may unintentionally overlook musical tunes that users genuinely enjoy. Therefore, the adoption of a rating-based fitness function appears to be a promising approach to increase the chances of providing music that aligns with the user's individual taste [9].

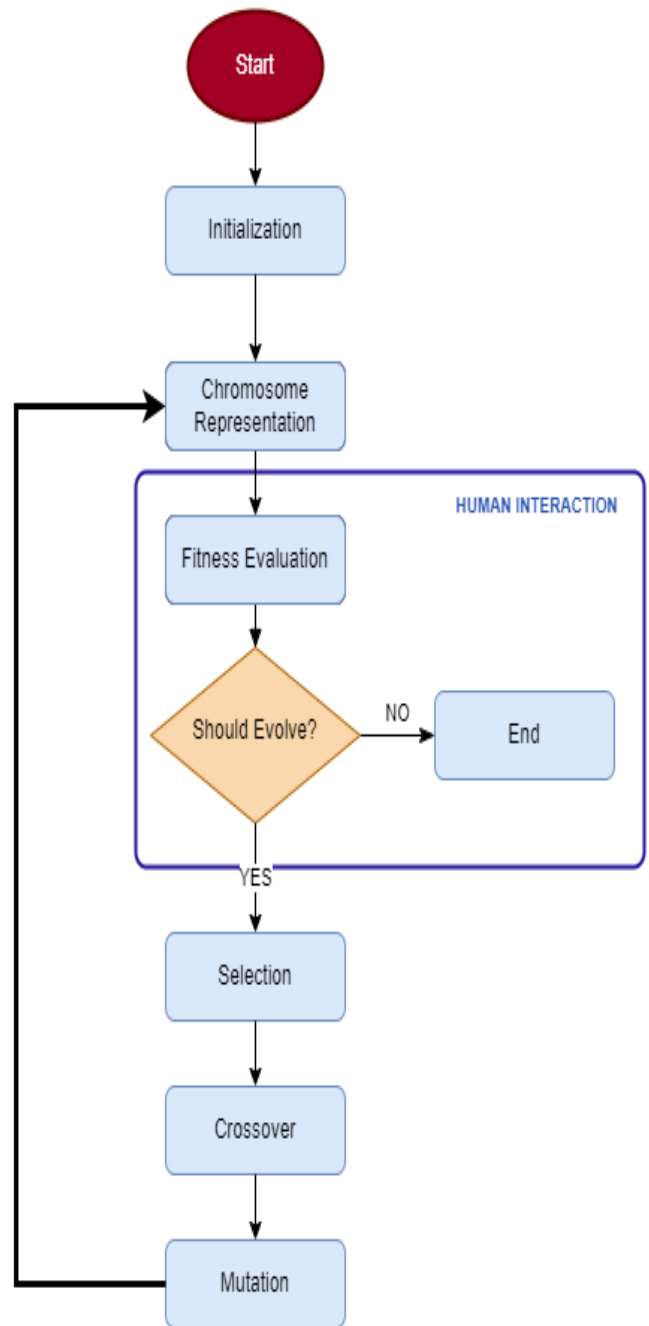


Figure 2: Flowchart for IGA in music generation

In a typical IGA scenario, as shown in Figure 2 through a visual flowchart, the process of generating a music melody involves active user participation. Users are typically presented with various potential solutions, and they subsequently evaluate these options based on subjective criteria. This iterative and user-centric approach allows for the refinement and optimization of the solution over successive iterations, ultimately leading to a music melody that aligns more closely with the user's preferences and objectives.

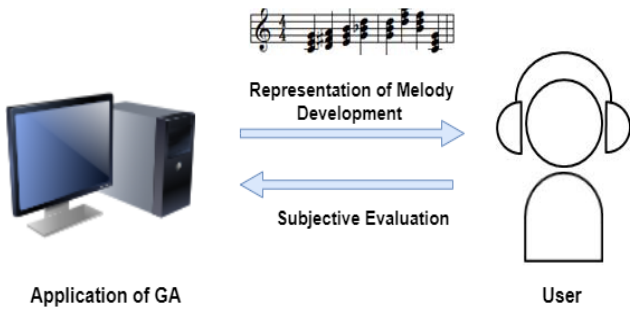


Figure 3: IGA schema for user-customized emotional music.

IGA involves active user participation. Users are typically presented with various potential solutions, and they subsequently evaluate these options based on subjective criteria. This iterative and user-centric approach allows for the refinement and optimization of the solution over successive iterations, ultimately leading to a music melody that aligns more closely with the user's preferences and objectives [4].

B. Construction of IGA system based on the proposed method

a) Encoding of the chromosomes

We employ an ArrayList data structure to represent each note, represented as a sequence of binaries, their durations, and intensities. Subsequently, we populate and manage all digits within this ArrayList. Which looks as follows:

11001	11101	...	00101	10011
-------	-------	-----	-------	-------

b) Representation of the melody

54	65	...	89	...	23	12	...	116
----	----	-----	----	-----	----	----	-----	-----

The number of bars and the number of notes represent musical notes encoded in binary, which are then converted into decimals. Alongside these decimal representations, various musical parameters such as scale, key, and others are considered as input. This collective input is used to generate a melody.

c) Termination conditions

The termination condition for this IGA is based on user satisfaction. The algorithm continues to generate and present solutions to the user until the user is satisfied with the quality of the solutions. User satisfaction is measured through user ratings.

d) Fitness function

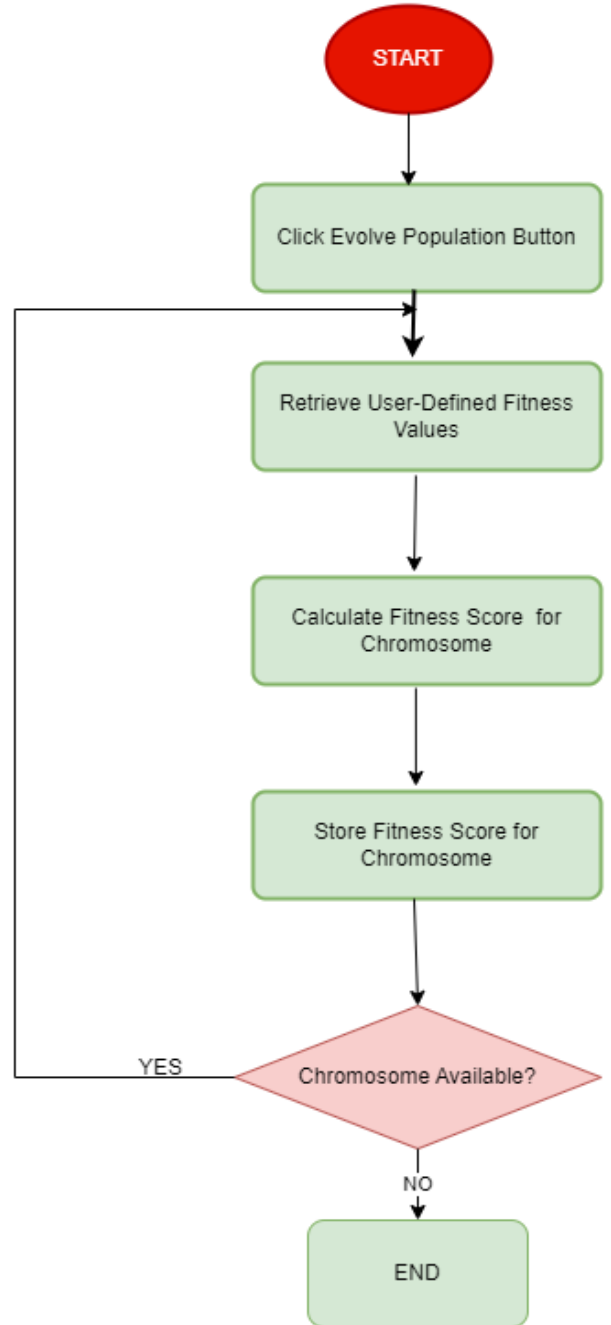


Figure 4: flowchart of fitness function

In an IGA, fitness calculation involves a user-centered approach. Candidate solutions, represented as chromosomes, are presented to the user, who evaluates them through a user interface and provides feedback based on their preferences and objectives. The higher the fitness score, the more desirable the solution is considered. Also, the time it takes for the user to listen to the music is taken into consideration, as such the longer you listen to the melody the higher the fitness value. The strength of IGAs lies in their adaptability to user preferences and problem-specific criteria, making them particularly useful in optimization scenarios where human judgment and subjective evaluation play a vital role in selecting the best solutions.

e) Selection functions

In our program's implementation, we experimented with two distinct selection methods: the roulette wheel and group selection, while also incorporating an elitism strategy. With elitism, the two most robust chromosomes are duplicated without modification into the new population, ensuring that the finest solutions discovered are preserved and not discarded during the evolutionary process.

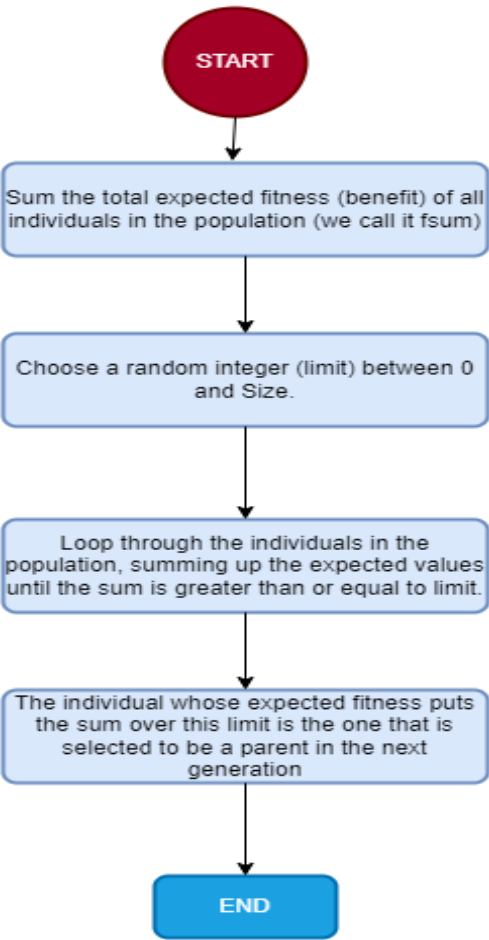


Figure 5: Roulette wheel selection

roulette wheel selection

The roulette-wheel method assigns portions of a circular wheel to individuals based on their fitness, and it is spun N times (with N as the population size). Selection continues until the mating pool is adequately filled, which is essential for generating new offspring [8]. Due to the small population size, relatively high selection rates, around 50%, are employed. This method is implemented as shown in Figure 5.

Elitism selection

We introduced this selection technique with the aim of enhancing the likelihood of favoring chromosomes with higher fitness values for guaranteed reproduction, while chromosomes possessing lower fitness values would have heavily reduced opportunities for reproduction [8]. To operationalize the elitism selection method, we employ an additional array named "indexes[Size]," where we store the indexes corresponding to the elements in the "chromosome_fitness[Size]" array.

chromosome_fitness				
0	2	3	4	
1	4	2	5	

Indexes				
0	2	3	4	
0	1	2	3	

The array "indexes" is arranged in descending order based on the fitness levels of the corresponding elements in the "chromosome_fitness" array. Consequently, the indexes associated with chromosomes possessing greater fitness values will be positioned at the start of the "indexes" array, while those with lower fitness will be situated towards the array's end.

indexes:				
0	2	3	4	
4	2	3	1	

Subsequently, we implement an elitism strategy that involves selecting an item from the group with the highest fitness (index 0). Following this, we proceed to the group with the second-highest fitness (index 1). This prioritizes

chromosomes with exceptional fitness levels, granting them a significantly higher likelihood of being chosen as parents for the upcoming generation. Below, you will find a visual representation in the form of a flowchart illustrating the intricacies of the group selection algorithm:

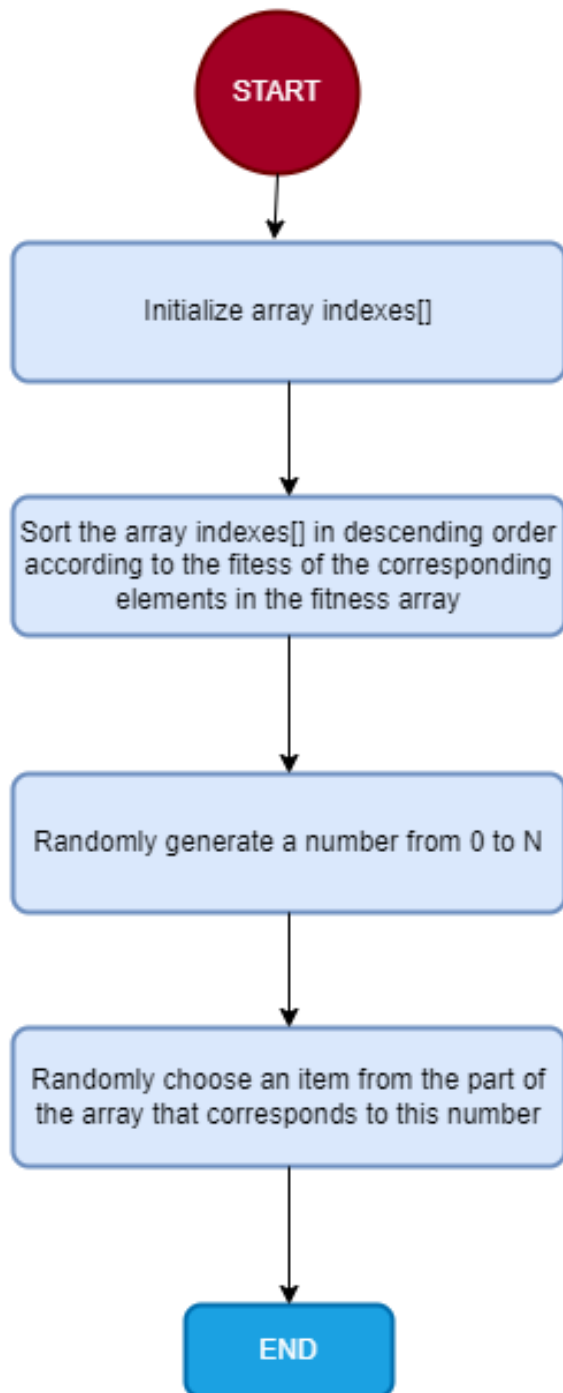


Figure 6: Elitism selection

f) Crossover

We experimented with both one-point and two-point crossover methods. Since there wasn't a significant difference in the outcomes obtained from these methods, we opted for the one-point crossover. In this approach, the crossover point is randomly determined by generating a number between 0 and $l-1$ (l = chromosome length). Crossover is performed with a certain probability: if the crossover probability is 100%, the entire new generation is created through crossover, while a probability of 0% results in the new generation being exact copies of chromosomes from the previous population. After testing the program with various values, we settled on a crossover rate of 60%, indicating that 60% of the new generation will be formed through crossover, while the remaining 50% will consist of copied chromosomes from the old generation.

g) Mutation

To ensure that the algorithm aligns with the user's preferences and efficiently converges to desired solutions, we execute the mutation process on every bit position within the chromosome, with a 0.15% probability for each bit.

V. RESULTS

The algorithm is written in Python, and Figure 7 shows its main interface. The system makes it possible to store interim results, which makes it easier to analyze how the algorithm is progressing. The phenotypes are saved and visualized using existing music frameworks.

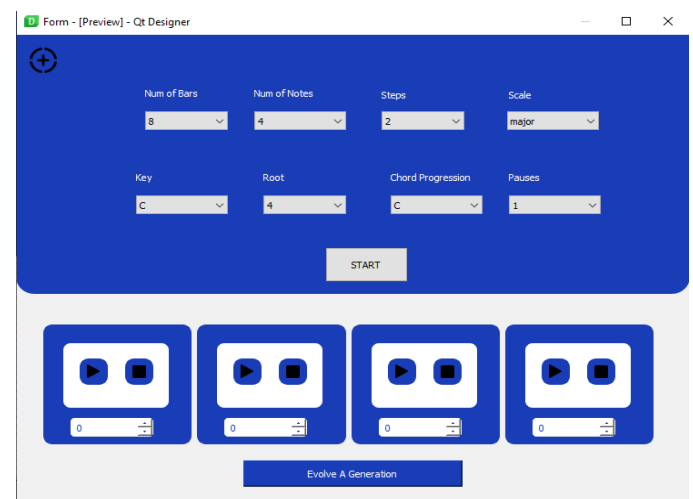


Figure 7: Interface of the Melodies Generator

In populations made up of three-note melodies, notable findings are often produced in a few rounds using a proposed mutation rate and a proposed crossover. The outcomes are even quicker for shorter melodies. A sample utilizing a population of just four notes is shown in Table 1 below.

Generation/ Chromosome	Generation 1	Generation 5
Chromosome 1	22, 45, 27, 37, 42, 41, 28, 40, 111, 34, 45, 38, 78, 25, 35, 25	53, 88, 83, 107, 52, 64, 50, 54, 45, 63, 11, 16, 119, 90, 95
Chromosome 2	89, 103, 56, 49, 120, 100, 60, 53, 39, 29, 9, 115, 34, 1, 58, 59	68, 22, 86, 123, 54, 65, 34, 111, 47, 63, 43, 28, 52, 54, 125
Chromosome 3	114, 63, 57, 64, 56, 53, 33, 59, 103, 8, 71, 7, 103, 9, 68, 57	53, 56, 3, 48, 71, 93, 64, 34, 4, 26, 33, 31, 52, 90, 21
Chromosome 4	119, 67, 88, 76, 49, 79, 49, 98, 11, 102, 80, 9, 9, 22, 0, 49	100, 22, 3, 107, 125, 29, 64, 34, 21, 26, 35, 31, 54, 94, 95

Table 1: Notation of the 2nd chromosome from the 1st and 5th generations.

In this basic application, once a pleasing melody is produced, the user is required to compose the beat. To guarantee the melody is consistent, there usually isn't a need for significant changes to the relative duration of the notes because the created sequences are normally quite simple. For many notes, such as quarters, it is reasonable to assume that their durations are equal. The choice of the proper beat, which is frequently hinted at by the music itself, is crucial. The tune represented by the second chromosome in the sixth generation of Figure 9 can be notated using standard musical notation with a 4/4-time signature by making the modifications as follows:



Figure 8: Representation of the 2nd chromosome from the 6th generation.

The five-noted pattern is engaging and motivating by itself. Additionally, simple, brief patterns like the one outlined above make wonderful foundations for creating complete naive tunes. These foundational themes can be easily expanded upon by musicians to produce an entire composition. For instance, as seen in Figure 9, these patterns can be easily expanded into a smoother melody:



Figure 9: A smoother naive melody created through IGA

In reality, the 16 iterations of the representation produced by the GA were used to create the whole melody shown in Figure 9, with only the notes inside the ellipsis being changed.

The challenge exists in instructing the system on how to create consistent, fully-formed compositions in the context of fully automated functions. Melodies can be compared to real discussions when questions and replies are exchanged while remaining focused on a single topic. Because of this, one may easily develop an initial theme by simply reacting to it and continuing in the same manner until they are satisfied with the results. Additionally, a large number of iterations become impracticable since the user, who plays a crucial role in the review process, may become weary. The melody shown in Figure 10 consists of four separate phrases, two of which duplicate the sequence produced by the GA while the other two share noticeable similarities, only requiring a few modifications to the original notes to achieve this similarity.

This goes to show that, the IGA achieves quick and efficient convergence, producing satisfying results with a minimal amount of interactions. Practically speaking, this is sufficient—at least for the time being.

VI. DISCUSSION

The analysis of preceding research looks into the different operations of genetic algorithms and their use in musical composition. One strategy involves using normalized information Distance as a fitness function in evolutionary algorithms, as indicated by literature [6], which is a novel approach. There are benefits and drawbacks to this strategy. The inconsistent way in which recombination techniques work during the course of evolution is a noteworthy problem, though. The most effective technique varies over time, and strategies that at first look promising may eventually lose their effectiveness. When choosing the best recombination technique for a genetic algorithm, this inconsistent performance might be a challenge because the best option may differ depending on the stage of evolution.

Another research [1] explains how artificial neural networks (ANN) can be used to automatically create music. In

this method, genetic algorithms are used to create music, and an artificial neural network (ANN) is tasked with selecting the most appropriate musical tune. A method for picking melodic music from a collection of created compositions is used by this automated system, which results in the automatic elimination of some musical compositions. As a result, it's possible that some of the music that is deleted contains melodies that the user likes. Additionally, Article [4] discusses the investigation of musical composition using genetic algorithms. In contrast to the usual method of controlling individual notes, this work distinguishes itself by using music generation through the intervention of chord progressions.

VII. CONCLUSION AND FUTURE WORK

This study has explored the interesting intersection between music composition and computer science, concentrating on the use of IGAs to produce music that matches user emotional preferences. The work has investigated the basic ideas and elements of IGAs, providing a comprehensive grasp of the GA, chromosomal representation, fitness functions, selection techniques, crossover, and mutation. Utilizing these methods, the research has shown how IGAs may effectively evolve musical compositions, placing a priority on user participation in the creative process. Additionally, the outcomes demonstrated how well this approach works at developing enjoyable and intriguing melodies through user engagement and feedback. The promise of IGAs in the area of music composition is highlighted by the algorithm's capacity to quickly converge on satisfying solutions with little involvement. An informative look into the future of AI-assisted music creation is offered by the study's practical application and utilization of fitness functions depending on user happiness. GA is an extremely useful tool for music composers because it makes the process of creating music quicker and easier. Professional composers may use it in the future to create movie tunes and other songs of other genres.

In our future work, we intend to combine our findings with those of other researchers [1], [6]. Our genetic algorithm will have this combination as its goal, enabling a thorough investigation of musical possibilities. Through the use of a sophisticated evaluator that can instantly identify the best melodies, this method seeks to promote diversity in the music generation. It takes care of the wide range of recombination tactics and makes sure that user preferences are taken into account throughout the evolutionary process.

Our current experiments concentrate on seeking melodies that can be deemed as an average representation of the target pieces. In future work, we aspire to introduce a component to the system that not only permits but encourages outliers. These alternative strategies will be tested with the expectation that they will contribute to a more refined and personalized automatic music composition system.

VIII. REFERENCES

- [1] Dong, L. 2023. Using deep learning and genetic algorithms for melody generation and optimization in music. *Soft Comput* 27, 17419–17433. <https://doi.org/10.1007/s00500-023-09135-3>
- [2] V. M. Marques, C. Reis and J. A. T. Machado. 2010. "Interactive Evolutionary Computation in Music," 2010 IEEE International Conference on Systems, Man and Cybernetics. Istanbul, Turkey. pp. 3501-3507. DOI:10.1109/ICSMC.2010.5642417.
- [3] Koga, S., & Fukumoto, M. 2014. A Creation of Music-Like Melody by Interactive Genetic Algorithm with User's Intervention. *Interacción*.
- [4] Fukumoto, M. 2014. Creation of Music Chord Progression Suited for User's Feelings Based on Interactive Genetic Algorithm. 2014 IIAI 3rd International Conference on Advanced Applied Informatics, pp 757-762.
- [5] Mitchell, M. 1998. *An Introduction to Genetic Algorithms*. Massachusetts: The MIT Press.
- [6] Alfonseca, Manuel & Cebrián, Manuel & De la Puente, Alfonso. 200). A simple genetic algorithm for music generation by means of algorithmic information theory. 3035-3042. 10.1109/CEC.2007.4424858.
- [7] ENGELBRECHT, A. P. 2007. "Genetic Programming." In *Computational Intelligence*. Chapter 10, pages 177-185. John Wiley & Sons, Ltd. ISBN: 9780470512517. DOI: <https://doi.org/10.1002/9780470512517.ch10>
- [8] Obitko, M. 2003. Czech Technical University (CTU). IV. Genetic Algorithm. Retrieved. <http://cs.felk.cvut.cz/~xobitko/ga/gaintro.html>
- [9] Satpute, Akanksha & Bajbalkar, Mayuri & Velankar, Makarand & Gurav, Saishwari & Abnave, Preeti. 2023. Soft Computing for Music Generation using Genetic Algorithm. *Journal of Soft Computing Paradigm*, 5. 11-21. 10.36548/jscp.2023.1.002.
- [10] Huang, D., & Xu, X. (2022). IGAOD: An online design framework for interactive genetic algorithms. *SoftwareX*, 19, 101205. <https://doi.org/10.1016/j.softx.2022.101205>
- [11] Carnovalini, Filippo & Roda, Antonio. (2020). Computational Creativity and Music Generation Systems: An Introduction to the State of the Art. *Frontiers in Artificial Intelligence*. 3. 10.3389/frai.2020.00014.